

P R O D U C T S P E C I F I C A T I O N S H E E T

DivertScan

Construction & Demolition Waste Compliance Platform

LEED v4 · LEED v4.1 · LEED v5

Every load, every ticket, every gram of carbon — verified.

DivertScan is the audit-ready compliance platform for construction & demolition waste — purpose-built for general contractors, sustainability consultants, owners, and recyclers working on LEED-certified projects. Capture every ticket, classify every load, hash every record, and produce GBCI-ready submittals without ever reconciling a binder of paper tickets the week before submittal.



HUMAN-VERIFIED COMPLIANCE — A NAMED VERIFIER SIGNS EVERY RECORD

OCR transcribes scale tickets. AI suggests a starting composition from debris photos. A trained human verifier then reviews the source materials, sets the final diversion values, and signs off — name and timestamp on every approval. The verifier is the responsible party for diversion findings; AI is a tool that speeds the clerical work. No AI-only path to "Verified" status. This is how DivertScan stays defensible at GBCI review.

Seconds

CAPTURE-TO-RECORD

100%

HUMAN-VERIFIED

SHA-256

HASH ON EVERY RECORD

v4 — v5

ALL BD+C, ID+C, NC

HOW IT WORKS

Capture → Verify → Report

Three stages. One audit trail. Zero manual reconciliation. The verifier is always the responsible party — the platform makes their job 10× faster.

01 C A P T U R E Pi reads the WI-127 indicator 24/7. iPad PWA captures tickets and photos. OCR auto-transcribes. AI suggests a starting debris composition.	02 V E R I F Y A trained human verifier reviews every record, sets the final diversion values, and signs off. SHA-256 hash chains the source data. No AI-only path to "Verified" status.	03 R E P O R T One-click submittal package for any LEED version. Monthly logs, end-of-project reports, embodied-carbon, hash manifest. Drop straight into LEED Online.
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WHY DIVERTSCAN

Built for the way GBCI actually reviews

Most compliance tools were designed for the version of LEED that existed when they shipped. DivertScan was designed for the GBCI reviewer reading your submittal — and the auditor who comes back six months later asking how you arrived at 87% diversion.

Defensible by design Every record is hashed (SHA-256) and signed by a named human verifier. Re-derive any number from source data in one command. No "trust us" math.	All three LEED versions Native support for v4 (MRp2 + MRc5), v4.1 (MRc5), and v5 (MRp1 + MRp2 + MRc5). One data model. No add-on modules. Switch versions mid-project.	Field-first, not office-first Captures at the scale — not after the dumpster pull. iPad PWA, Pi-side hardware, driver loadbook URLs. Works offline. Syncs deltas only.
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COMPLIANCE COVERAGE

LEED Coverage Matrix

DivertScan natively supports every C&D-related credit and prerequisite across LEED v4, v4.1, and v5 — from the original v4 MRp2 planning prereq through the v5 zero-waste and embodied-carbon prereqs. No add-on modules. No separate report templates.

VERSION	CREDIT / PREREQUISITE	DIVERTSCAN SUPPORT
v4 BD+C	MRp2 — C&D Waste Management Planning (prereq)	Auto-generated WMP from project setup. Includes diversion targets, hauler list, recycling facilities.
v4 BD+C	MRc5 — C&D Waste Management (2 pts)	Diversion calculated by weight (lb/tons) or volume. ADC and waste-to-energy correctly excluded per credit rules.
v4.1 BD+C	MRc5 — C&D Waste Management (prereq folded into credit)	Total-waste-generated approach supported. Path 1 (diversion %) and Path 2 (lbs/sf) both calculable.
v5 BD+C	MRp1 — Planning for Zero Waste Operations (prereq)	Zero-waste operations plan auto-assembled from project setup, hauler agreements, and material flows.
v5 BD+C	MRp2 — Quantify and Assess Embodied Carbon (prereq)	CO ₂ e calculations per material stream (concrete, metal, wood, drywall, asphalt) with documented emission factors.
v5 BD+C	MRc5 — C&D Waste Diversion (2 pts)	Diversion narrative auto-generated with material breakdown, end markets, and tonnage. PDF + XLSX export.
All	GBCI LEED Online submittal package	One-click export: WMP, monthly logs, end-of-project reports, weighmaster certificates, SHA-256 hash manifest.

THE RULES THAT GET AUDITED

Source-Separation & Reporting

Every credit-determining rule in LEED MRc5 and the v5 MRp1/MRp2 prereqs is encoded into DivertScan's data model. Below is how the platform handles each of the most-audited requirements — the ones most projects get wrong.

Material streams defined by end-market	Each diverted load is tagged by destination facility AND end-use (recycle, reuse, fuel, ADC, landfill). LEED defines a stream by where the material ends up — wood-to-fuel and wood-to-reuse count separately when tracked separately. Supports v4 (4 streams for 2 pts), v4.1 (2–3 streams), and v5 (optional but documented).
Commingled = 1 stream rule, enforced	Per LEED rules, commingled counts as ONE stream unless the facility provides project-specific per-material weights. DivertScan enforces this in the calculation — a commingled load can't inflate your stream count by accident.
RCI-certified path supported	For commingled material, both paths supported: project-specific diversion rates from the sorting facility (must be weighed & sorted, not visual), or facility-average from a regulated facility. RCI status stored per facility and flagged on reports for v4.1+ projects.
ADC counted as waste	Alternative Daily Cover (ADC) is automatically excluded from diversion totals and counted as waste, per LEED rules. Material destination tagged as "ADC" is bucketed correctly without manual reclassification.
Wood-to-biofuel counted as diversion	Wood diverted to biofuel counts as diversion per LEED rules. Other waste-to-energy paths NOT counted as diversion (unless European Waste Framework Directive exception applies). Platform tracks end-use to apply the correct rule.
Excluded materials auto-removed	Excavated soil, land-clearing debris, on-site reused materials, and hazardous materials are automatically excluded from generation calculations. Tag a ticket as any of these and it's removed from the math but preserved in the audit log.
Demolition tracked separately	For v4.1 Option 2 and similar paths, demolition waste must be tracked separately from new construction. Every ticket is tagged with project phase (demolition / construction / sitework) — separate totals on every report.

Weight OR volume — consistent throughout	Calculations can be by weight or volume, but must be consistent across the whole project per LEED rules. DivertScan locks the unit at project setup. Mixed-unit imports flagged for resolution before any report can be generated.
lbs/sf threshold (v4.1 Path 2)	For "Generate less than 7.5 lbs/sf (BD+C)" or "2.5 lbs/sf (C&S)", DivertScan auto-calculates project waste density from total weight ÷ certified floor area. Threshold compliance shown on dashboard in real-time as construction progresses.
Facility regulation status	Each hauler/sorting facility stores: regulatory authority (city/county/state), facility type (recycler / commingled MRF / landfill / WTE / reuse), and RCI certification status with expiry date. Reports surface any facility status issues before submittal.
Monthly diversion tracking	Per LEED reference-guide recommendations, diversion rate and active streams are tracked monthly — not just at end of project. Dashboard shows month-over-month trend. Auto-emailed monthly reports keep the project team aligned.
Final waste management report	End-of-project report includes total waste generated, diversion rate, material streams with weights and end markets, hauler list, facility list with regulatory status, narrative explaining strategy, and SHA-256 hash manifest. Ready to upload.

S T A G E 0 1

Capture

How tickets, photos, and weights get into DivertScan in the field. Every AI output below is human-verified before commit.

Pi-side scale capture	Live weight reads from any truck-scale indicator (WI-127, AvWT, Rice Lake, Cardinal, others) over RS-232 or Ethernet. Auto-classifies gross / tare / noise. No driver intervention. Runs 24/7 even when the iPad is in the office.
GPT-4o Vision OCR	Photograph any scale ticket — handwritten, dot-matrix, thermal, faded — and DivertScan extracts ticket #, gross, tare, net, hauler, project, date in seconds. Every result queued for human review before the record is finalized.
Debris photo pre-sort (AI-assisted)	Snap a photo of the load. The platform suggests a starting % composition (concrete, wood, drywall, metal, mixed C&D) as a sorting aid. A trained human verifier reviews the photos and sets the final material percentages.
EXIF time-match	Drop a folder of phone photos and DivertScan auto-binds them to the correct scale ticket using EXIF timestamp ± 2 minutes. Human verifier confirms every match in the review queue before commit.
Batch OCR with weight cross-check	Scan a stack of 50 tickets at once. The platform OCRs all of them, cross-validates gross-tare=net against the indicator log, and surfaces any mismatch > 2 lb to the human verifier.
Walk-in inbox	Drag-and-drop tickets from any source — emailed PDFs, phone photos, fax scans, hauler portal exports. Everything funnels into one review queue.
iPad-first PWA	Installs to home screen. Works in any modern browser. Field crews capture tickets even without cellular — writes are buffered locally in IndexedDB (Vanessa Protocol) and auto-sync the moment connectivity returns. Bounded-latency Supabase calls (Sullivan Engine) keep the UI responsive on bad Wi-Fi.
Driver loadbook URLs	Each driver gets a unique URL to view their own loads, daily totals, and ticket history. No login. No app install. Just a bookmark on their phone. Reduces phone calls to the office at 4:30 PM.

S T A G E 0 2

Verify

How DivertScan makes every record audit-defensible — through cryptographic hashing AND mandatory human verification.

Human verifier signs every record	Mandatory workflow. OCR transcribes; AI suggests a debris pre-sort. A trained human verifier reviews the source materials, sets the final values, and signs off before the record commits to the LEED submittal. The verifier is the responsible party. No record reaches "Verified" without human sign-off, enforced at the database layer.
SHA-256 audit hashing	Every ticket, photo, weight reading, and edit produces a SHA-256 hash chained to the previous record. GBCI reviewers can re-derive any hash from source data — proving nothing changed since capture.
Orphan-resolution drawer	Scale readings without a matching ticket land in an orphan queue. Resolve one-at-a-time or batch-merge to the correct ticket with one tap.
Fuzzy driver-name matching	OCR'd driver names match against your driver list even with misspellings, abbreviations, or initials. Reduces 80%+ of "unknown driver" tickets without manual cleanup.
Photo review queue	AI-flagged photos (low confidence, unusual composition, ticket discrepancies) surface in a queue for mandatory human review. No photo reaches the LEED record without a verifier signing off.
Time-matched chain of custody	Every weight reading, photo, and ticket carries a UTC timestamp. Reconstruct the full sequence of events for any load at any time.
Tamper-evident edits	Edits never overwrite — they create a new record with a delta. Original is preserved with its hash. No "history rewrite" possible. Every edit logs who, when, and what changed.
Login + access audit log	Every admin and client portal login logged with timestamp, IP, and device. Supports compliance audits and security reviews. Default retention: 7 years.

STAGE 03

Report

How DivertScan turns human-verified, hash-chained data into submittal-ready documentation.

One-click LEED submittal package	Generate the complete MRp1 / MRp2 / MRc5 submittal for any LEED version in seconds, not days. PDF, XLSX, and JSON formats included. Drop straight into LEED Online.
Embodied-carbon (CO₂e) calculations	Per-material emission factors applied automatically. Output suits LEED v5 MRp2 narrative requirements. Editable factors for jurisdiction-specific overrides. Documented sources for every factor.
Material breakdown	Concrete, metal, wood, drywall, mixed C&D, asphalt, cardboard — broken out by ton, %, and end market. Both weight and volume views.
Diversion narrative in plain English	Auto-generated narrative explaining the project's diversion strategy, hauler relationships, and end markets — ready to paste into LEED Online or adapt to project-specific language.
Monthly / quarterly / project-end reports	Schedule auto-emailed reports to the GC, owner, consultant, and AHJ on the cadence each one wants. Custom recipients per project.
Multi-project portfolio rollup	Consultants managing 5+ projects see one combined view: aggregate tonnage, diversion %, CO ₂ e avoided, and comparison vs. benchmarks across the portfolio.
QBO invoice export	Bill haulers, owners, or back-charges directly into QuickBooks Online from the same dataset that drives the LEED reports.
Hash manifest for GBCI reviewers	Every report ships with a manifest of SHA-256 hashes. Reviewers verify integrity in one command (openssl/python script included). Speeds GBCI review turnaround.

FOR EVERYONE IN THE CHAIN

Stakeholders & Access

Each role gets the view tailored to their job. Same data, different lens.

Client portal (per-project)	Owners and GCs log in to see only their projects. Real-time tonnage, diversion %, recent tickets, and on-demand reports.
LEED consultant portfolio view	Consultants managing multiple projects see all in one workspace, with cross-project comparisons and combined exports.
Hauler management	Add, edit, and deactivate haulers. Track which haulers contribute the most divertable material — useful for negotiating better contracts.
Driver loadbook (read-only)	Drivers get a unique URL to see their own loads. No login, no app install — just a bookmark on their phone.
Role-based admin permissions	Operations staff edit tickets. Sustainability staff finalize reports. Owners read only. All controlled centrally.
Samsara fleet integration	Pull telematics data from Samsara to cross-validate driver IDs, truck IDs, and route history against scale tickets.
Shareable demo link	Send a prospect a self-contained demo URL with a 60-second auto-tour. No login required.

UNDER THE HOOD

Platform & Architecture

iPad-first PWA	Installs to home screen. Works in any browser. No App Store approval cycles, no MDM friction.
Raspberry Pi scale-side capture	Plug into any truck-scale indicator with RS-232 or TCP/IP. Local SQLite buffer means the Pi keeps recording even if the Supabase cloud is briefly unreachable, then re-syncs automatically when connectivity returns.
Sullivan Engine — 5-second bounded latency	Every Supabase call is wrapped in a 5-second AbortController timeout. Bad Wi-Fi can no longer freeze the UI waiting on a TCP handshake that will never finish. Repeated timeouts auto-detect a degraded mode so the app can back off intelligently instead of hammering the network.

Vanessa Protocol — local-first sync queue	If Sullivan reports a timeout, ticket writes are queued in IndexedDB on the device. Field crews keep capturing in the basement of a hospital project; the queue drains automatically when connectivity returns or on a 30-second timer. A "pending" pill in the header shows the queue depth at a glance.
SHA-256 hash chain	Every ticket, weight reading, and edit hashes the prior record. Reviewers can re-derive any hash from source data to prove nothing changed since capture.
PostgreSQL / Supabase backend	Encrypted at rest (AES-256). TLS 1.3 in transit. US-East data residency. Row-level security on per-client portal access.
GPT-4o Vision (OCR + pre-sort)	Modern vision model reads tickets and suggests a starting debris composition. Human verifier sets the final values that go into the LEED record.

SECURITY & COMPLIANCE

Defensible by design

Built on a security model that gives auditors what they actually want: cryptographic proof of integrity and clear access boundaries.

Encrypted at rest + in transit	AES-256 on the database. TLS 1.3 on the wire. No exception.
Per-record SHA-256 hash	Every record produces a deterministic hash. Reviewers can re-derive from source data — proving data integrity without trusting the platform.
Row-level security	Clients only see their own data. Drivers only see their own loads. Enforced at the database layer, not the app layer.
Audit log retention for the life of the project	Every login, edit, and report generation logged with timestamp, user, and SHA-256 hash. Records are preserved as long as the project remains active.
SOC 2 Type II path	Architected for SOC 2 readiness. Penetration testing planned as customer base grows. Reports will be available under NDA once completed.
GDPR / CCPA compliant	Right-to-delete and right-to-export available on request for any client.

P R I C I N G

Simple, predictable, project-based

Pilot — 1 project, 30 days	Free pilot to prove the platform on your next LEED submittal. No credit card. Outcome: a real, signed-off submittal package you keep — even if you don't continue.
Standard — per active project / month	Unlimited tickets, photos, OCR, AI pre-sort, human verification, and reports. No per-user fee. No add-on modules.
Portfolio — sustainability consultants	Unlimited projects under one organization. Cross-project rollup view included. Volume discounts at 10+ active projects.
Enterprise — GCs & owner-developers	Multi-team access, SSO, dedicated support, custom data residency.
Onboarding included	Every pilot includes 2 hours of setup support — connecting your scale, importing your driver list, configuring projects.

**Migration from
spreadsheets / Green
Badger / Topia**

CSV import for existing project history. Hash manifest re-derivable on imported records.

**Stop reconciling spreadsheets
the week before submittal.**

See it run on a live project

divertscan.com/divertscan-demo.html

T A L K T O R O B E R T

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